

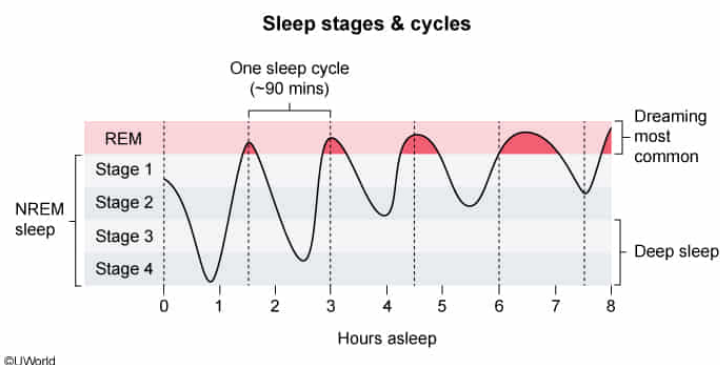
Which of the following observations disconfirms the hypothesis that REM sleep is critical for memory consolidation?

- ✔ ☒ A. Individuals taking antidepressant drugs that profoundly suppress REM sleep show no learning disruptions. [84%]
- ☐ B. Individuals deprived of REM sleep one night tend to exhibit REM rebound, more REM sleep than usual the next night. [1%]
- ☐ C. Individuals deprived of sleep tend to perform worse on memory tasks than those who are not deprived of sleep. [12%]
- ☐ D. Studies suggest that dreaming occurs during both REM and NREM sleep but is more common during REM sleep. [1%]

Correct

83% answered correctly

Explanation:



Sleep is divided into non-rapid eye movement (NREM) sleep (stages 1-4) and rapid eye movement (REM) sleep. **Brain activity** during **REM** sleep resembles an **awake, alert state** but with the body functionally **paralyzed**, except for the muscles controlling respiration and eye movement.

The hypothesis that memories are consolidated (processed and stored) during REM sleep has not been supported by data. While it appears that sleep is involved in the consolidation of certain types of

memory, research has yet to determine which types are consolidated during specific sleep stages.

REM sleep is suppressed by several classes of antidepressants, yet these drugs do not appear to disrupt learning or memory processes. Therefore, this observation *disconfirms* the hypothesis that REM sleep is critical for memory consolidation.

(Choice B) Individuals who are deprived of REM sleep one night exhibit more REM sleep than usual the next night (ie, REM rebound).

However, this observation does not relate to memory function, so it does not disconfirm the hypothesis that REM sleep is critical for memory consolidation.

(Choice C) Sleep-deprived individuals tend to perform worse on memory tasks than those who are not sleep deprived. However, this observation does not elucidate whether REM or NREM sleep is critical to memory consolidation, so it does not disconfirm this hypothesis.

(Choice D) Studies suggest that dreaming occurs during both REM and NREM sleep but is more common during REM sleep. However, this observation does not relate to memory function, so it does not disconfirm the hypothesis that REM sleep is critical for memory consolidation.

Educational objective:

REM sleep is characterized by rapid closed-eye movements, body paralysis, and brain activity resembling a waking state. Dreaming is most common during REM sleep but can occur in other stages as well. REM sleep deprivation for one night results in REM rebound (more REM sleep than usual the next night).

Time spent:0 sec
Subject:
Behavioral Sciences

QID:402011
System:
Sensation, Perception, and Consciousness